

Task

Instructions

Design, implement, and present a fully functional data-driven web API with database integration (**Python + Django + PostgreSQL**)

GenAI Usage

This task intentionally allows you to use **GenAI** (e.g. debugging, planning, proof-reading etc). You must declare all tools used, purposes, and attach examples of exported conversation logs as supplementary material. **Expectation:** Creative application of **GenAI**, such as exploring high-level alternatives and reimagining the design of cutting-edge solutions.

Some examples of GenAI Usage

- 1) Use AI to search online for datasets in areas you want to explore. Ask AI to suggest relevant datasets, data sources, or domain-specific repositories.
- 2) Provide a subset of your dataset to AI and ask it to help you create a script to import and store the data in your database.

You need to include a GenAI declaration, analysis, and conversation logs as an appendix in the report.

Deliverables: 4 in total

1: Code

- 1) Local GitHub repository with commit history demonstrating consistent version control.
- 2) Includes a clear README.md describing setup instructions and project overview.

2: API Documentation

- 1) Generated or manually written documentation (e.g. Swagger UI, Postman, Markdown).
- 2) Clearly describes all available endpoints, parameters, and response formats.
- 3) Includes example requests and expected responses (JSON).
- 4) Documents authentication process and error codes.
- 5) Must be referenced in the README.md file as a PDF file (if generated as an HTML file, then convert to PDF).

3: Technical Report (5 pages – appendices not counted)

- 1) Concise written report outlining key design and architectural choices. I need it to contain your personal choices for the technology stack.
- 2) Justifies the decision of programming language, frameworks, and databases.
- 3) Reflects on challenges, testing approach, and lessons learned.
- 4) Includes limitations and potential areas for improvement or future development.
- 5) **Absolutely Important:** contains a **GenAI** declaration and thoughtful analysis of its usage. You must attach examples of exported conversation logs as supplementary material.

4: Presentation Slides

This will include dedicated sections covering:

- 1) Version control practices
- 2) API documentation
- 3) Technical report highlights
- 4) All deliverables (code, API docs, report)

Do not make the slides text heavy. Instead put extra information in the notes below

Data Sources

You must use publicly available datasets for this task. Below are some suggested platforms where you can access suitable data:

- 1) **Kaggle** (<https://www.kaggle.com/datasets>) – Large collection of datasets across various domains, including books, weather, health, and more.
- 2) **data.gov.uk** (<https://www.data.gov.uk/>) – UK government open data portal with datasets on transport, health, education, and public services.
- 3) **OpenWeatherMap** (<https://openweathermap.org/api>) – Weather data API that can be used to collect historical and current weather information.
- 4) **Google Dataset Search** (<https://datasetsearch.research.google.com/>) – Search engine for publicly available datasets.
- 5) **UCI Machine Learning Repository** (<https://archive.ics.uci.edu/>) – Datasets commonly used for machine learning and data analysis projects.
- 6) **NHS Digital** (<https://digital.nhs.uk/data>) – Health and care data from the UK National Health Service.

Absolutely minimum requirements

I need more than this, however, as an absolute **minimum requirement**, please make sure you satisfy the following requirements:

- 1) Implement at least one data model involving all four elements of CRUD (Create, Read, Update and Delete) functionality linked to a database.
- 2) Design is based on RESTful API
- 3) Include at least four API endpoints accessible via HTTP requests
- 4) Handle user inputs and return appropriate JSON responses
- 5) Use correct status/error codes, according to industry convention
- 6) Be demonstrable via local execution, web hosting (e.g. PythonAnywhere), or a Model Context Protocol (MCP) server

E.g. you could include 10 API endpoints but if you don't use the correct status/error codes then it's not met the requirement